DISRUPTED SPIRAL GALAXY

ESO 510-G13

CATALOG NUMBER ESO 510-G13

DISTANCE 150 million light-years

DIAMETER 105,000 light-years

MAGNITUDE 13.3

HYDRA

Despite being referred to only by a number rather than a name (its long designation comes from the European Southern Observatory's catalog), ESO 510-G13 is one of the most intriguing galaxies in the sky. It is an edge-on spiral with a clear dust lane marking its central plane. The dust lane has an obvious twist.

The most obvious explanation for the kink is that ESO 510-G13 has had a close encounter or collision with another galaxy in its recent past. Some astronomers have suggested that the collision is still going on, and the dust lane is the "ghost" of a galaxy that ESO 510-G13 has swallowed—as seen in the active galaxy Centaurus A (see p.322). Alternatively, the disk might have been warped by the gravity of a nearby galaxy. The galaxy responsible might be a small neighbor or a more distant but larger member of the same group. As their techniques and instruments improve, astronomers are finding this kind of distortion is common in spirals, although it often shows up more in the distribution of gas than in the stars, so it is usually most obvious at radio wavelengths. Our near neighbor M31 (see pp.312–313) has such a distortion, and the Milky Way seems to have one, too, perhaps caused by interaction with its own family of smaller neighbors.

WARPED DISK
The bright core of ESO 510-G13 silhouettes the galaxy's warped dust lane in this image. The blue glow on the right is a huge area of bright young stars—evidence, perhaps, of a collision in the galaxy's recent history.

SB0 BARRED SPIRAL GALAXY

NGC 6782

CATALOG NUMBER NGC 6782

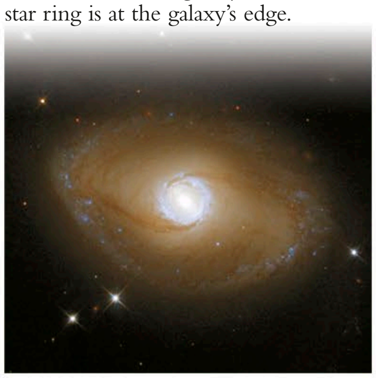
DISTANCE 183 million light-years

DIAMETER 82,000 light-years

MAGNITUDE 12.7

PAVO

The Hubble Space Telescope imaged the apparently normal barred spiral galaxy NGC 6782 in 2001. Using ultraviolet detectors, it studied the pattern of the galaxy's hottest material. The image (see below) showed, in pale blue, two rings of stars so brilliant and hot that they emit most of their light as ultraviolet. The inner ring lies in the galaxy's bar and could have been ignited by tidal forces between the bar and the rest of the galaxy. The outer star ring is at the galaxy's edge.



ULTRAVIOLET STAR RINGS

DISRUPTED SPIRAL GALAXIES

The Mice

CATALOG NUMBER NGC 4676

DISTANCE 300 million light-years

DIAMETER 300,000 light-years

MAGNITUDE 14.7

COMA BERENICES

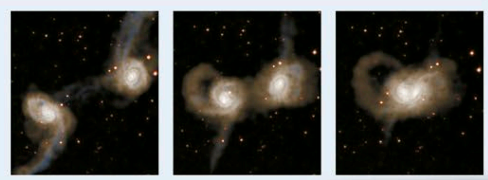
The object classified as NGC 4676 is in fact a pair of colliding galaxies—called the Mice because they appear to have white bodies and long, narrow tails. As with the Antennae Galaxies (see p.317), the long streamers are the result of the spiral arms "unwinding" during the collision—though in this case, one of the arms lies edge-on to us and so appears to be long and straight despite being strongly curved away from us. Knots of bright blue stars in the streamers and the main bodies of the galaxies show where bursts of star formation are taking place. Computer simulations of the collision (see panel, right) suggest that the galaxies are now separating after a closest approach 160 million years ago.

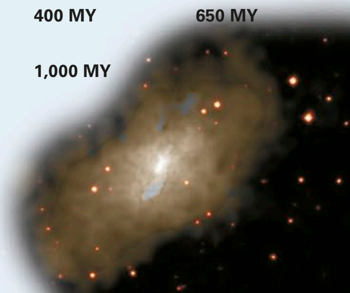
HIDDEN EXTENT
Image processing allows astronomers to amplify faint light from the outlying parts of the Mice, revealing their true shape and extent.

EXPLORING SPACE

SIMULATING GALAXY COLLISIONS

The great challenge for astronomers studying colliding galaxies is that they can only ever see one stage in a story that unfolds over millions of years. Fortunately, today's supercomputers can help speed things up. By building "model" galaxies with simplified star clouds, gas, dust, and dark matter and then smashing them into each other in a computer, astronomers can measure how gravity affects the fate of the galaxies.





SPIRAL COLLISION SIMULATION
This computer simulation shows two spiral galaxies interacting and merging to form a large, irregular galaxy. Time is measured in millions of years (My).

DESTINED TO UNITE
Although currently moving apart from a close encounter, the Mice are gravitationally locked together and doomed eventually to merge, perhaps resulting in the formation of a new giant elliptical galaxy.